



150 years of the Pula Observatory



HRVATSKA
ZAJEDNICA
TEHNIČKE
KULTURE
1946 - 2021

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Introduction

This collection of photographs and graphics from the history of the Hydrographic Institute (since 1969 an office) of the Austro-Hungarian Navy, i.e. its Observatory, is not intended as a historical or scientific publication. The intention behind this document is to convey the knowledge that the members of the Astronomical Society "Istra" have gained through their activities in a simple and comprehensible, popularizing way to everyone who is interested in it.

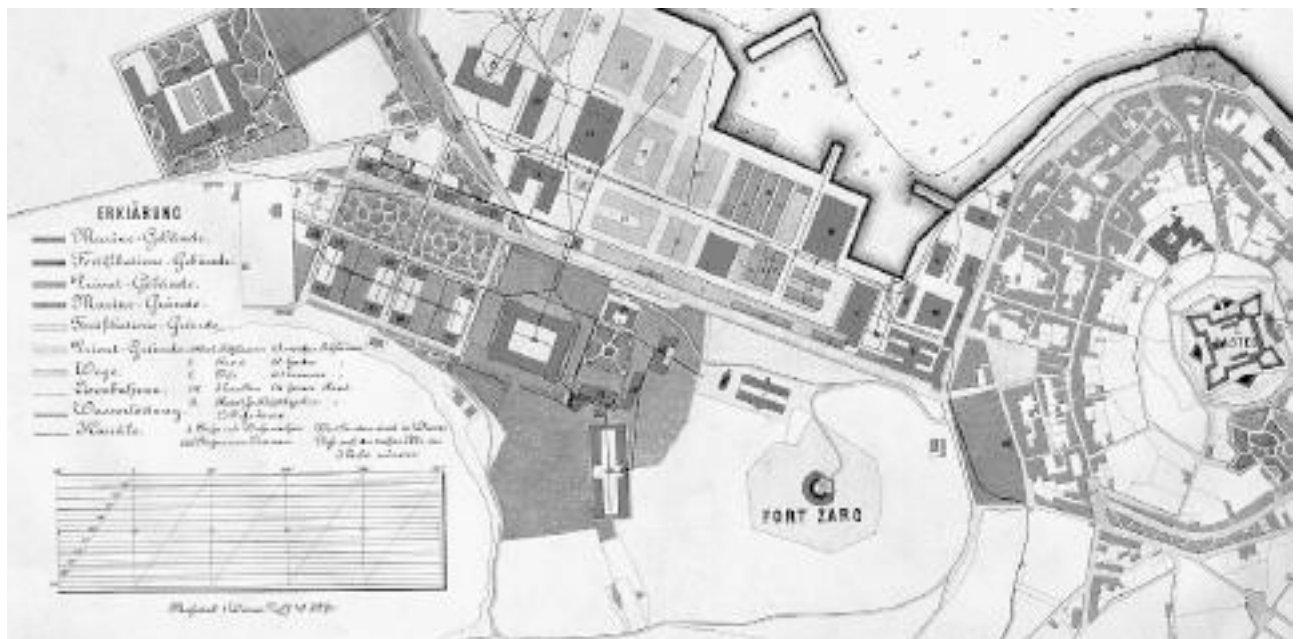
The historical facts that are presented along with individual pictures are mostly taken from the book "Observatory in Pula: 1871-1991. (materials for the history of Istra)" which was published in 1991 as a special edition by "Histria Historica" and can be found in public libraries throughout Croatia. Part of the facts related to the discoveries of Johan Palisa and his contemporaries were taken from the Minor Planet Center service. The history that includes the activities of the Astronomical Society "Istra" Pula has been transmitted first-hand, that is, from the archives of the Association.

The source of the historical photos is the Historical and Maritime Museum of Istria, which in its archive, in addition to photographic material, also has a significant collection of postcards (mostly from the era of the Austro-Hungarian Monarchy), one of which was chosen as the front page. The remaining photos and copies are part of the archive of the Astronomical Society "Istra" Pula or provided from the collections of individual members of the Association (Edi Puh and Luciano Gržinić, images taken from aircraft were provided by Igor Karasi). In cases where it is not possible to determine the source of the image or data, and we consider it necessary, the source is listed separately.

The Croatian Association of Technical Culture is celebrating its 75th anniversary this year, and since the Astronomical Society "Istra" Pula is part of the Association, a separate part of this publication is dedicated to that anniversary as well.

prepared by, for
The Croatian Association of Technical Culture Pula
Astronomical Society "Istra" Pula
Damir Šegon



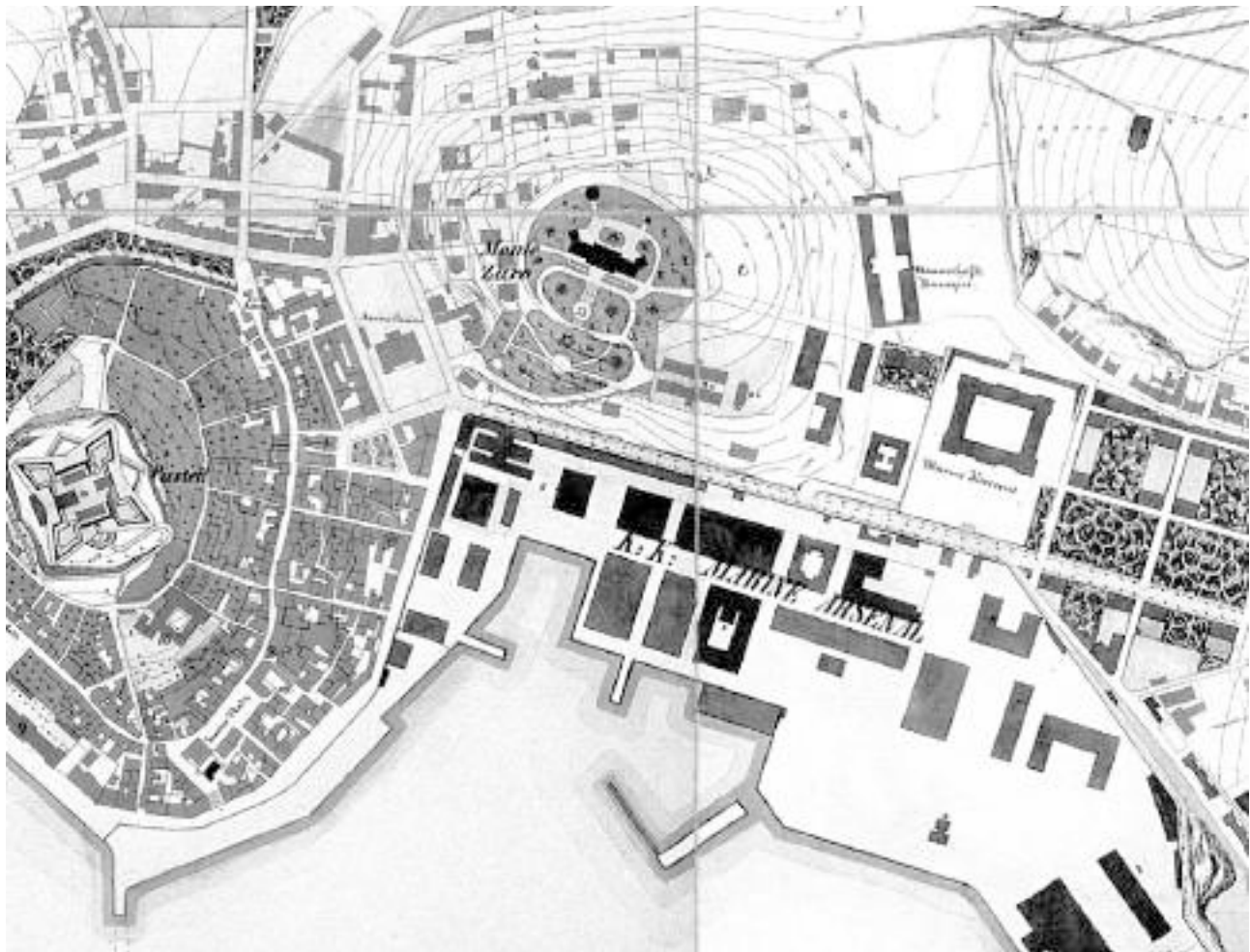


After Pula became one of the ports of the Austrian Navy (K.K. Kriegsmarine), planned construction of fortifications began. For this purpose, in 1853, the construction of a fort began on the city hill of Zaro (Monte Zaro), whose cannons could reach almost the entire Gulf of Pula.

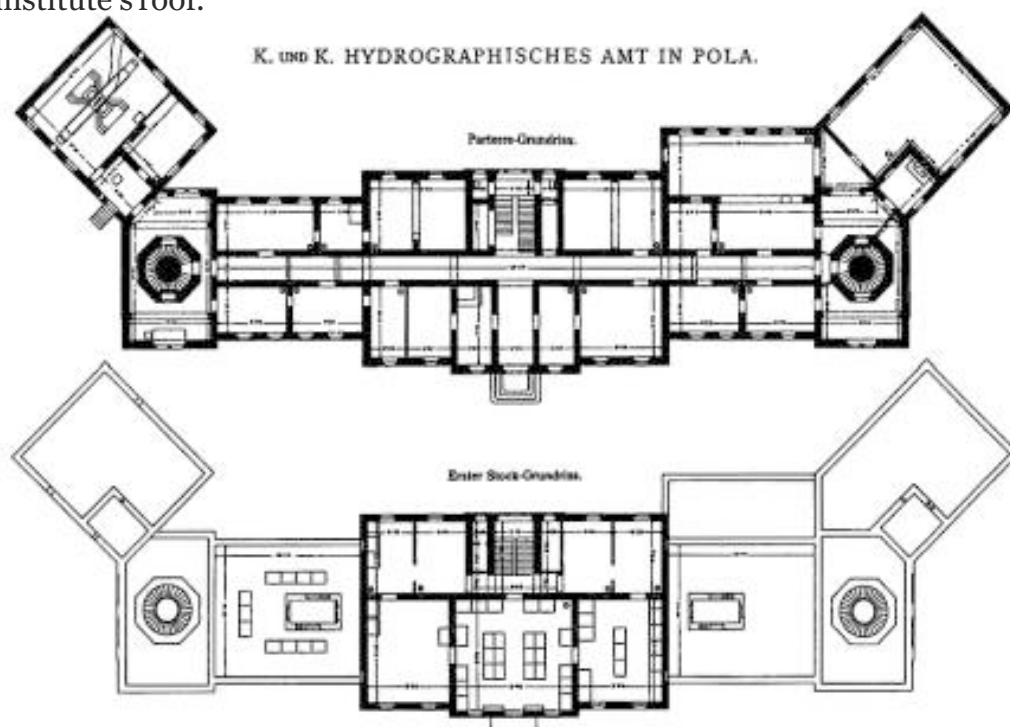


With the improvement of defense construction and technology, primarily the increase in range and precision of cannon projectiles, and the construction of new forts further from the city center, the Zaro fort lost its original significance.

With the growth of the Pula war port, there was also a need for the establishment of an institution that will be a scientific service for the Navy, primarily to enable the most precise navigation as possible.



It was decided to demolish the Zaro fort and build a modern building of the Hydrographic Institute with all the necessary facilities on that site. This process was started in 1869. The location for the new building was no accident. It was visible from almost the entire harbor, which gave the ships in it an unobstructed view of the device for giving an accurate time signal, located on institute's roof.



K. UND K. HYDROGRAPHISCHES AMT IN PULA.



The Naval Observatory in Pula was founded in 1869 as one of the departments of the Imperial and Royal Hydrographic Institute (office) of the Navy (K. und K. Hydrographisches Amt).

Already in 1871, on the site of the former fort, we see the new building of the Hydrographic Office, which, among other things, was also intended as an observatory with the task of maintaining an accurate time service using astronomical methods. It should be noted that in addition to the tasks prescribed by the Statute, the officers of the Observatory department, as military personnel, mostly engaged in other astronomical observations on their own initiative.





Panorama of Pula from the period between 1870 - 1880, apparently taken from the city's Monte Ghirio hill.

The observatory dominates the view not only over the Pula harbor but also much further.





Panorama of Pula taken from an airplane in 2015.

On the sketch, the remaining parts of the Observatory that were not damaged in the Allied bombing in 1944 are marked on the left.



Panorama of the port of Pula from around 1900. After the construction of the new building, a park was built, which became one of the most beautiful in Pula. Navy ships from their scientific missions around the world brought diverse flora with which the park was arranged.





Panorama of Pula taken from an airplane in 2015.

On the sketch, the remaining parts of the Observatory that were not damaged in the Allied bombing in 1944 are marked on the left.



One of the rare shots of the Observatory where all its domes can be seen. The only recording where you can clearly see the proportions of the large meridian dome, from where the measurements of the time of the stars passing through the meridian were carried out.



In 1877, a monument to Vice-Admiral Wilhelm von Tegetthoff (who became famous when in 1866, as the commander of the Austrian war fleet defeated the Italian fleet near the island of Vis) was erected in front of the Hydrographic Office.

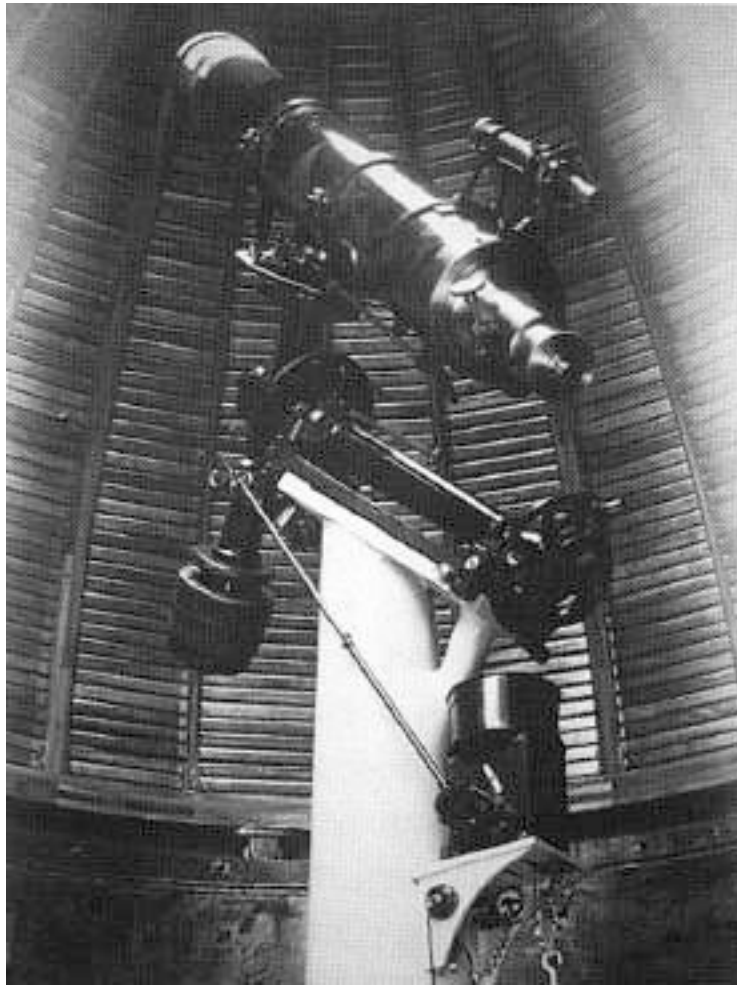
View of the central part of the Observatory from the back.





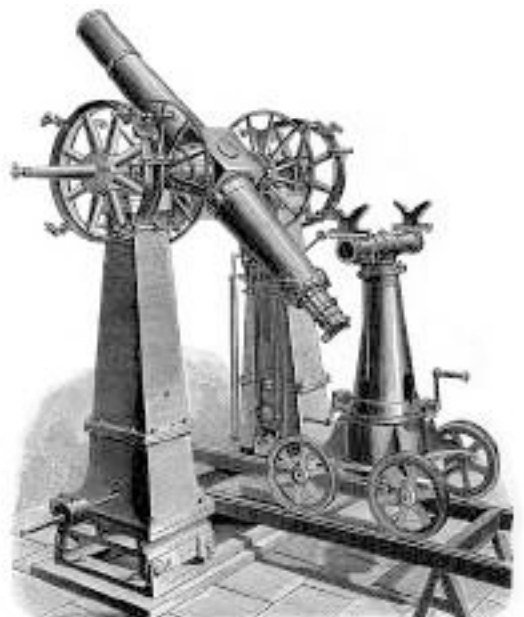
View of the central part of the Observatory from the back. Above: the north dome (survived the Allied bombing in 1944) Below: the central part, on which you can see the meteorological instruments for measuring wind speed and direction.

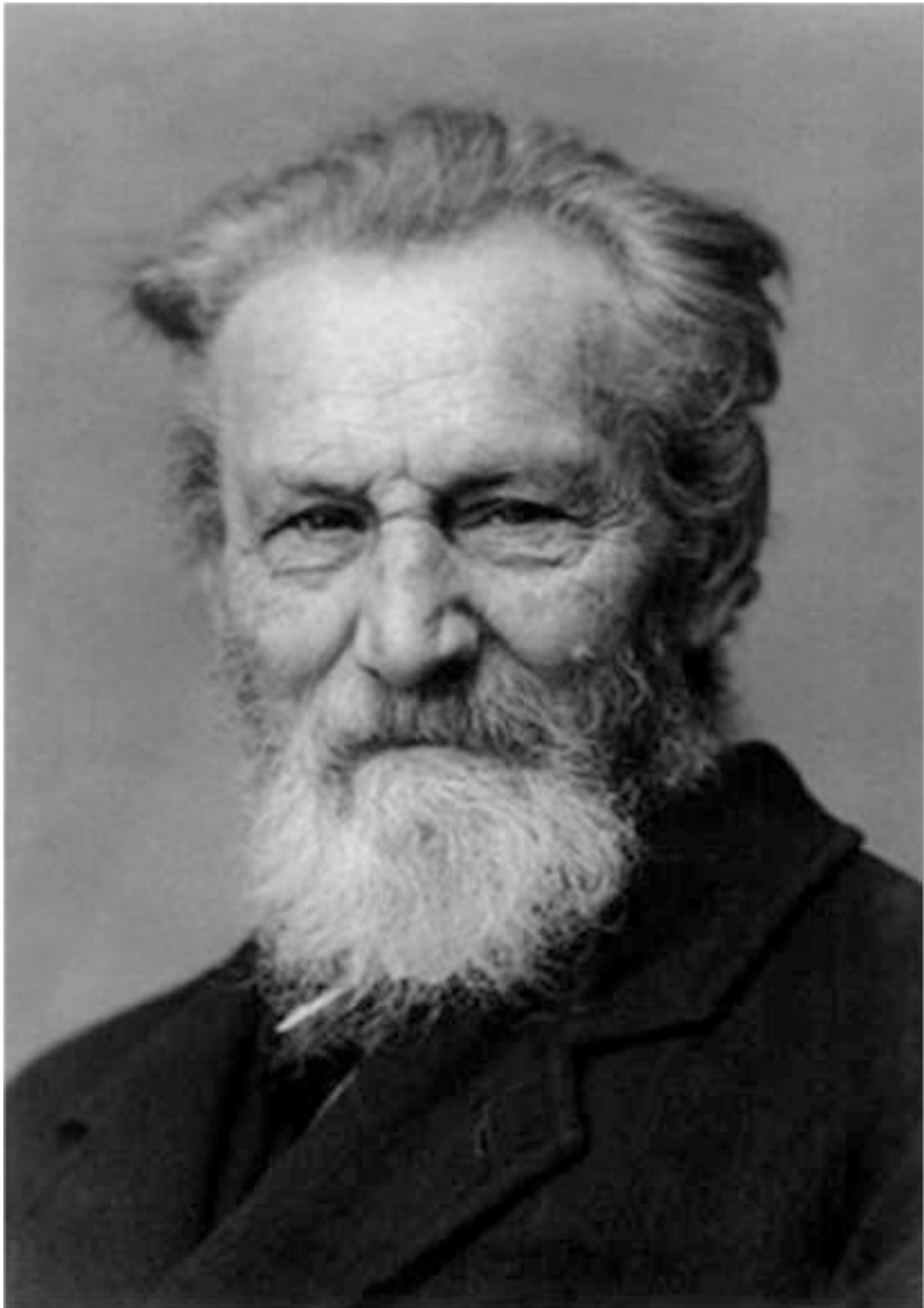




Some of the astronomical instruments used at the Observatory. Above: "Steinheil" refractor, an objective made by Reinfelder und Herdtel. It had a lens diameter of 160mm with a focal length of 2100mm, and was used for observing comets and drawing sunspots.

Bottom left: Brachyt-type telescope, 300mm aperture, which made it the largest reflector telescope of its kind in the world. Bottom right: "Troughton & Simms" meridian circle. It had a lens with a diameter of 160 mm, with a focal length of 2105 mm. The measuring circles were 900 mm in diameter with four microscopes on each side. It was used for the needs of the accurate time service, as well as for determining the position of the stars needed in the creation of star catalogs.





The most famous astronomer from the Observatory in Pula, thanks to whose work it is forever inscribed in history, is certainly Johann Palisa.

He continued his studies in mathematics and astronomy in practice as an assistant at the Vienna Observatory in 1870, where he worked on determining the positions of stars and planets and drawing sunspots. At the beginning of 1871, he was transferred to the observatory in Geneva, and a few months later to the Naval Observatory in Pula, where he became the Head of the Astronomical Department at the age of only 24.

While conducting observations of known small planets, he was also looking for new ones, so after a few years his efforts were rewarded: on March 18, 1874, Palisa managed to discover his first small planet. As the first asteroid discovered from Austria, Palisa assigns it the name Austria.

Palisa discovered a total of 28 asteroids and one comet (C/1879Q1) from the Pula Observatory. How much this meant on a global scale is also shown by the statistics of discoveries from that period, from which it can be clearly concluded how efficient Palisa was in his observations.

With a total of 123 discovered asteroids, he still has the largest number of visual discoveries to date.

otkrivač	otkrića
Palisa, J.	28
Peters, C. H. F.	22
Watson, J. C.	7
Borrelly, A.	6
Perrotin, J.	5
Henry, P. P.	5
Henry, P. M.	5
Coggia, J.	3
Knorre, V.	2
Schulhof, L.	1
Cottenot, P.	1

observatorij	otkrića
Pola	28
Clinton	22
Paris	10
Marseilles	10
Ann Arbor	6
Toulouse	5
Berlin	2
Peking	1
Vienna	1

godina	Palisa	ostatak svijeta	Palisa %
1874	3	3	50%
1875	6	11	35%
1876	0	12	0%
1877	1	9	10%
1878	3	9	25%
1879	10	10	50%
1880	5	3	63%

Statistics of the discovery of asteroids in the world, during Palisa's work in Pula (in 1880 he was promoted at the observatory in Vienna).

Palisa's asteroid legacy is intertwined with several famous names. On the occasion of his visit to Pula on April 8, 1875, Emperor Franz Joseph I personally named the two newly discovered small planets; (142) Polana and (143) Adria. If we add the asteroid (183) Istria, there are three celestial bodies in the universe that bear the names of our geographical terms (Polana - Puležanka, Adria - Jadran and Istria - Istra).

As a sign of recognition by the International Astronomical Union, next to the crater on the Moon, Palisa has two asteroids associated with its name: Palisana (914) is an asteroid named after Johann Palisa, while the asteroid Pawona (1152) is named in honor of two top scientists of that era, Johann Palisa and Max Wolf.



Space probes on their way to the outer planets of the solar system inevitably had to pass near one of the asteroids. Since Palisa discovered mostly large asteroids, some of them were recorded by space probes - such as asteroids 243 Ida and 253 Mathilde.

To everyone's surprise, images sent by the Galileo space probe in 1993 showed that asteroid 243 Ida has its own natural satellite, named Dactyl.

As 243 Ida was photographed in great detail, it became the first asteroid to be named after geological features: the three regions were named Palisa (after the discoverer Johann Palisa) and Vienna and Pola (after the observatories where Palisa worked and searched for asteroids).

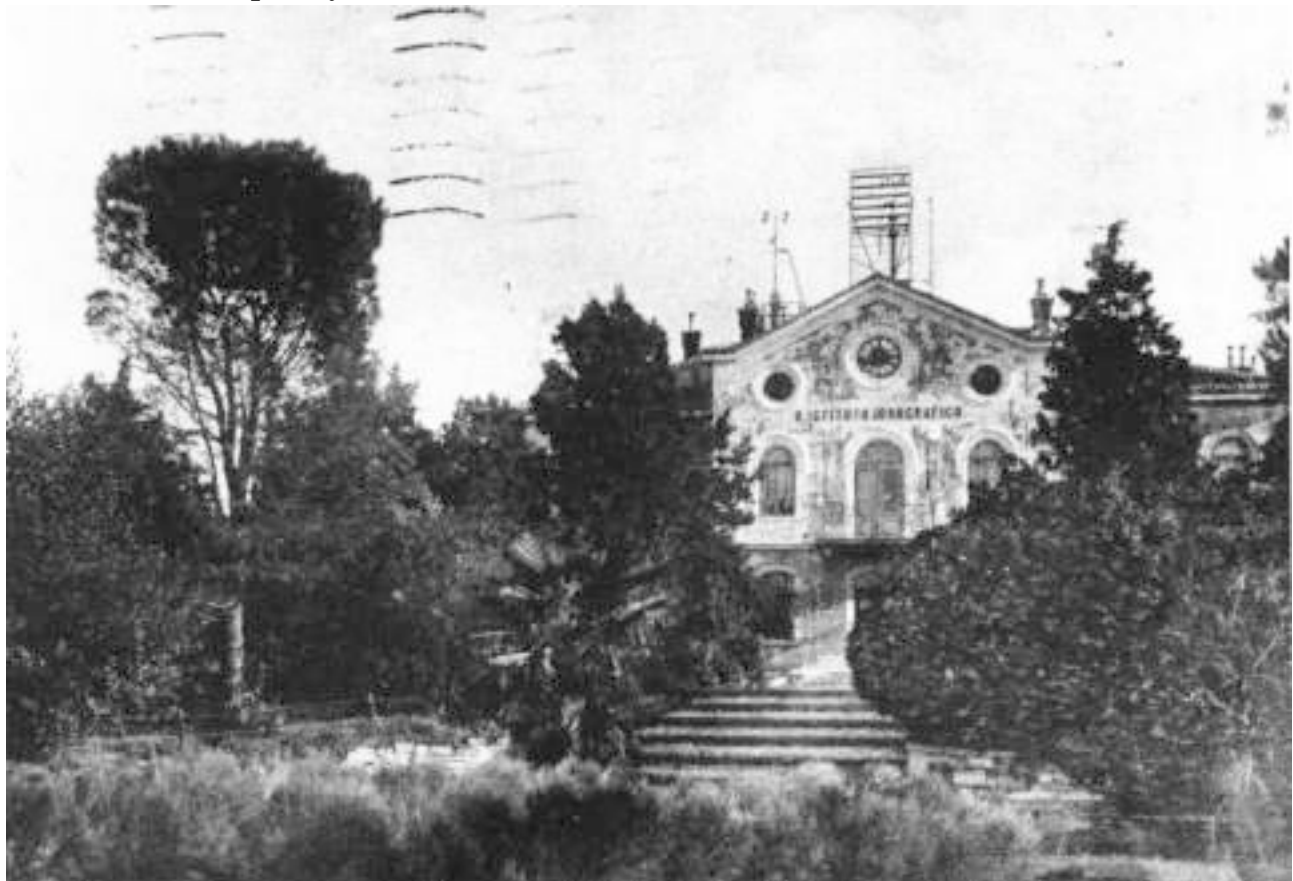


After the collapse of Austria-Hungary and the Italian military occupation of Pula on November 5, 1918, activities at the Observatory cease. The instrumentation and library were largely moved to Trieste and other Italian cities, and the monument to Admiral Tegetthoff, on whose initiative the Hydrographic Office was established, was also removed.





The fate of the Observatory after the collapse of the Austro-Hungarian monarchy and the occupation of Istria by Italian military force coincided with the general stagnation of the city of Pula. The Pula Hydrographic Office was renamed the Italian Royal Hydrographic Institute, with the status of a branch of the central Italian hydrographic institution. Meteorological and geomagnetic observations were carried out within that branch, while astronomical observations completely died out.





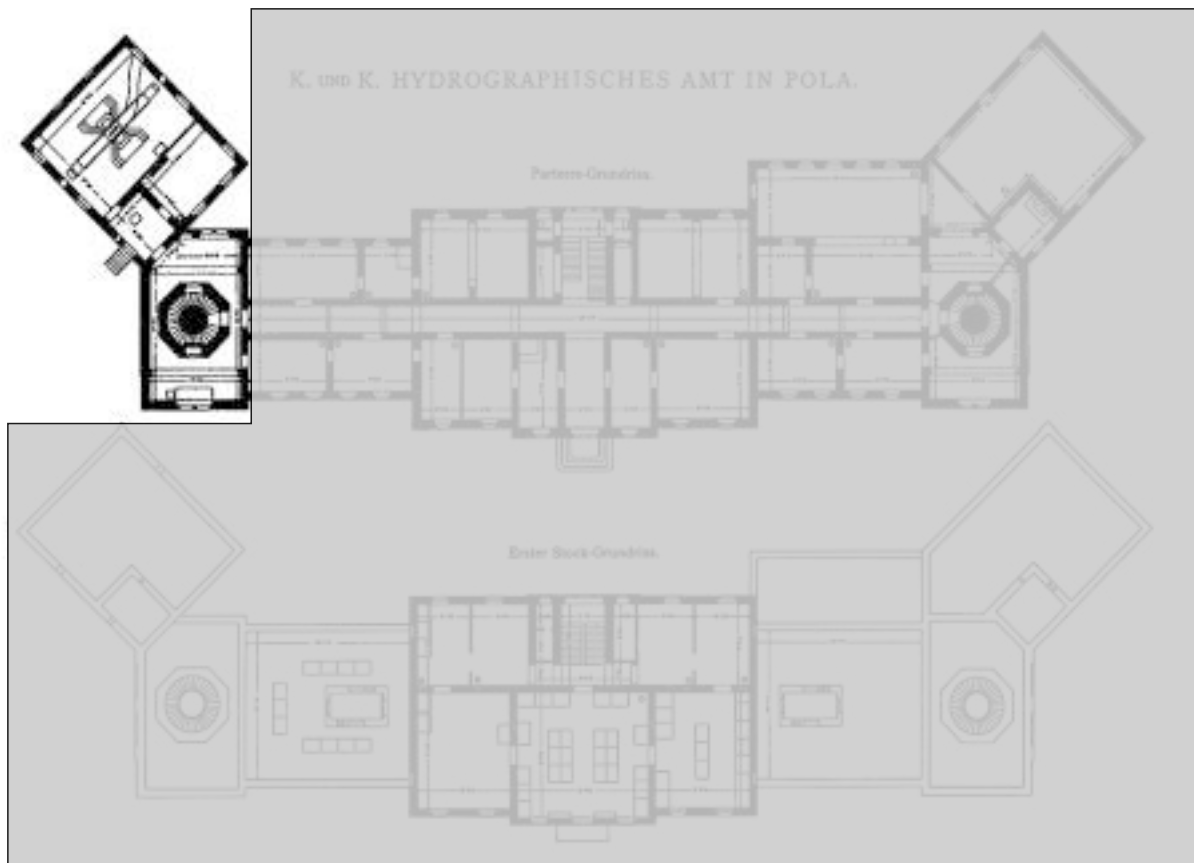
After the withdrawal of the Italian forces, during 1944, the Allied aviation repeatedly bombarded Pula. As the Hydrographic Office and the observatory are still listed as military facilities on the maps, Allied bombs destroy the buildings to the ground. On the footage from the bomber raid, the Observatory can still be seen intact (above) and either directly hit or already destroyed by bombs (below).





A picture of the remains of the Observatory after the Allied bombing (above). Only part of the northeast wing was partially preserved, while the rest was completely destroyed.

From the original floor plans of the building, it can be approximately estimated that only one small porch and only one dome (below) remained of the original surface of the building.



23. June 1946.



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The Croatian Association of Technical Culture is an umbrella organization of technical culture in the Republic of Croatia that works on the development and promotion of technical culture, and in terms of its structure and scope it is unique in Europe.

The association is the historical successor of the Republican Commission "Technique and Sport" at the Main Board of the Croatian Physical Culture Association, founded on June 23, 1946 in Zagreb. Academic Andrija Mohorovičić, a respected geophysicist and scientist in the field of seismology and meteorology, became the first president of the Republican Commission "Technology and Sport". The association is the legal successor of the social organization of the National Technical Society of Croatia founded on April 17, 1948 in Zagreb, which was renamed the Croatian Community of Technical Culture on July 3, 1992 and took over all its rights, obligations and assets. Istria founded its branch of the Community after the withdrawal of the allied forces in 1947.

The Croatian Association of Technical Culture, in accordance with the Law on Technical Culture, in the drafting of which it participated, bases its activities on the principles of responsibility, expertise and adaptability in working with users, on innovation, openness, respect for diversity and equality of participation.

In addition to the Nikola Tesla Technical Museum in Zagreb, the Croatian Technical Culture Association brings together 18 national associations of technical culture, 19 county and 35 city associations of technical culture.

Implementing programs of the public needs of the Republic of Croatia in technical culture, today the Croatian community of technical culture is especially focused on maintaining programs of extracurricular and extracurricular activities for children and young people, in which the practical application of knowledge develops their motor skills, independence, creativity, responsibility, critical reflection and observation and they are directed to the selection of production, professional and engineering occupations. The most famous programs of lifelong education and popularization of technical culture that the Croatian Community of Technical Culture has been conducting for years are: Competition of young technicians, Summer School of Technical Activities at the National Center of Technical Culture in Kraljevica, Model League, Robokup, Festival of Technical Culture, professional development of teachers and mentors of technical culture and magazine of ABC techniques.

In 2021, the Croatian community of technical culture is celebrating 75 years of continuous and organized activity. On the occasion of the 75th anniversary, a monograph was prepared in which valuable historical material, continuity and the most important achievements from 1946 to 2021 were processed. The importance of the Croatian Technical Culture Association was also recognized by the Croatian National Bank, which issued a HRK 25.00 coin with the mark of its 75th anniversary, which will probably be the last HRK 25.00 coin before the changeover to the euro.

Most of the programs are held by the Croatian Association of Technical Culture throughout the Republic of Croatia with the participation of its members, and some in cooperation with the Education Agency and with the support of the Ministry of Science and Education. New collaborations that began in 2020 and 2021 with the University of Rijeka, with the faculties of the University of Zagreb and with the University itself, with the Institute of History, the Agency for Vocational and Adult Education, the Adriatic Aerospace Association, the Croatian Fire Brigade and many others, will enable the Croatian community of technical culture to expand its future activities, to the benefit of the entire society.



The partially preserved part of the northeastern wing was renovated during 1947/48. and the meteorological station of the Yugoslav Navy is located in it. In 1949, this station was taken over by the newly established Directorate of the Hydrometeorological Service under the Government of the Republic of Croatia, today's State Hydrometeorological Institute of the Republic of Croatia.



The restoration of the remains of the Observatory was not entirely successful, mainly for technological reasons. Meteorological measurements of temperature, atmospheric pressure and air humidity were carried out from the measuring house in the fenced area in front of the building, while the heliograph and anemometer were placed on the auxiliary scaffolding on the south side of the dome.

The building housed two families of meteorologists, who continued their meteorological observations from Monte Zaro. It is important to point out that meteorological observations are still made from the same location today. Today, the meteorological station of the State Hydrometeorological Institute on Monte Zaro has the status of a climatological station.



During the restoration of the building, two domes were reconstructed, the classic and the meridian dome. The skeleton that can be seen in the picture immediately after the bombing was used to restore the original shape of the dome, covered with wood on the inside and galvanized sheet on the outside of the dome.

The mechanism for turning the classic dome in the form of a hemisphere remained in usable condition, as well as partially the mechanism for its opening. The opening of the dome was repaired in such a way that by opening it, it was possible to observe objects up to only 30 degrees from the horizon, so that the largest and best part of the sky would be inaccessible to the observer.

The impacts during the bombardment distorted the turret itself to a certain extent, so that the force required to turn it is not equal but depends on the positions in which the turret opening is located.

However, since the dome was not used for any observations (it did not open or turn at that time), there was no access to it, but the stairs led only to the terrace.

The meridian dome was seriously damaged, and its mechanism is unusable today, even though it exists in its original form. Only the covering part of the dome was reconstructed to protect it from further deterioration. The gear shafts are twisted from the force of the impact, the chain transmissions are no longer complete, so its reconstruction is still waiting for better days.



Inside the main dome, the original functionality in terms of opening and turning has been fully preserved to this day. The axial ball bearing on which the dome rests was cast during the construction of the Observatory in the Arsenal, where such bearings were produced for the purpose of gun turrets on warships of the Austro-Hungarian Monarchy. According to our knowledge, it is the largest bearing of its kind that is still in operation.





The Meridian dome today (above). The basic elements of the sliding door mechanism continue to withstand the test of time. The interior of the meridian dome (below) shows obvious signs of deterioration due to the long-term impossibility of quality maintenance of the specific structure of the dome.



27. December 1973.

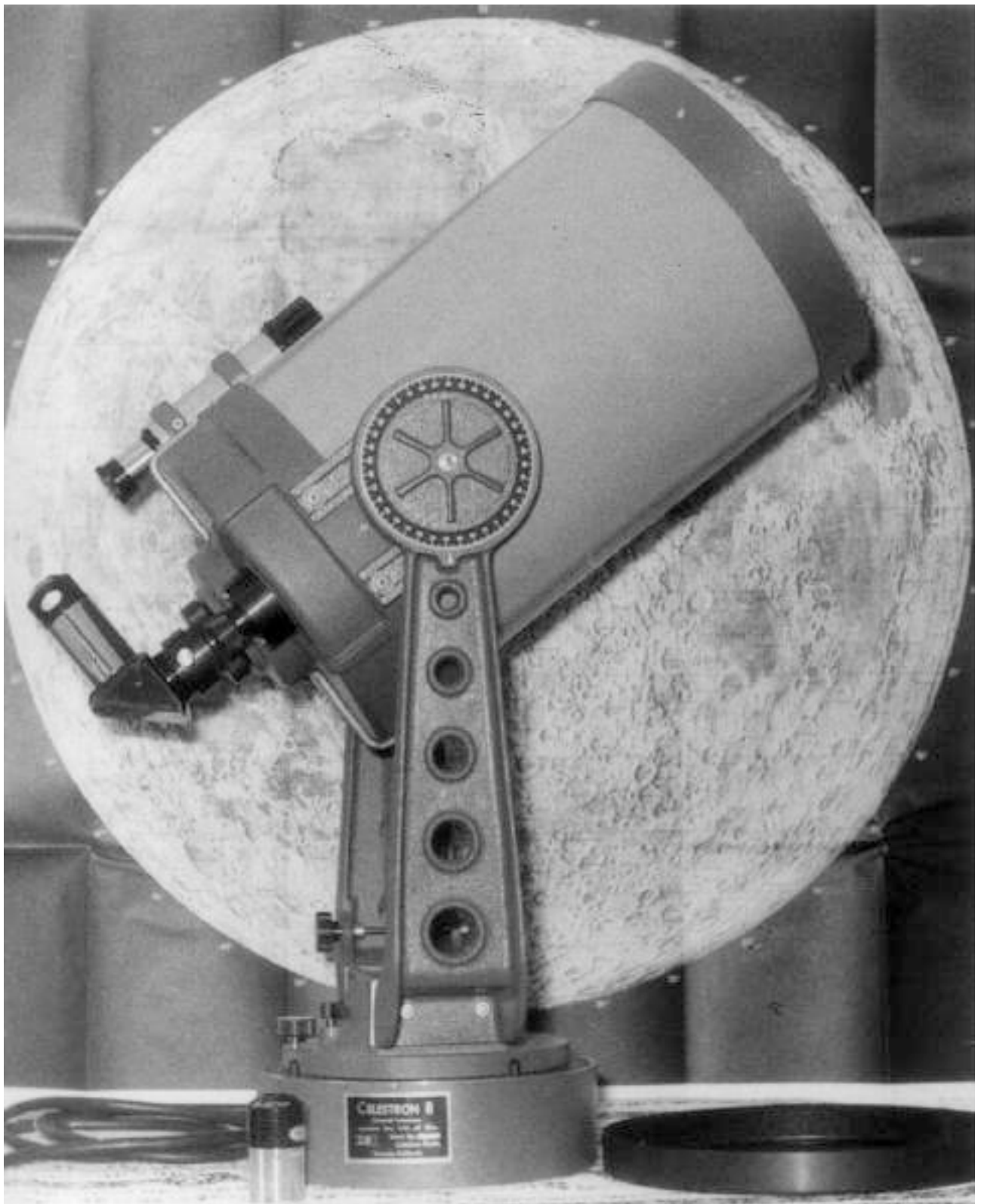


Astronomical Society "Istra" Pula was founded at the time of research endeavors in the field of astronomy and astronautics, at a time when the thirst for knowledge and the acquisition of new skills was difficult to quench due to the lack of literature, resources and organization.

The primary task of the society at that time was the popularization of astronomy and other natural sciences, and as such it was available to all ages in the form of lectures and observation of celestial phenomena. A very good connection with schools from the very beginning resulted in a mass response of elementary school students to the optional subject of astronomy, so from the end of the 70s of the last century, excellent results were recorded at the competitions of the "Science for the Young" movement, organized by the then National Technique Association, followed by today's Technical Association cultures.

A group photo of part of the membership of the Astronomical Society "Istra" Pula, from a visit to the Bazovica observatory in 1975. The imagination of the society's observatory began to take off...





The telescope, the basic astronomical instrument, was not as easy available as it is today. Purchasing power, in addition to high prices on the market, as well as restrictions on imports by the state, were the main problems. However, in 1975, the Astronomical Society "Istra" Pula managed to collect funds and buy the telescope "Celestron 8", which is still used today for the purpose of observing celestial bodies.

For several months, the instrument was exhibited in the window of the then "National Technique" and caught the eyes of passers-by, and aroused the curiosity of many fellow citizens. Observations at that time were held on the terraces of buildings in the city center, or not far from the city, since the level of light pollution was much lower than today.

Although the Astronomical Society "Istra" Pula in its beginnings had no direct contact points with the Observatory, its heritage was cherished. Thus, on the occasion of the hundredth anniversary of the discovery of the asteroid "Istra", an appropriate lecture was held, as well as an exhibition of works and telescopes made by members of the society themselves. Hundreds of people attended the anniversary celebration.



In cooperation with the "Arena" philatelic society, a philatelic exhibition with topics from astronomy and astronautics was prepared, and a commemorative envelope and commemorative badge were issued (it was a time when badges were very popular).



The request of the Astronomical Society "Istra" in Pula for the use of the Observatory building was already sent in 1974 to the Republic Hydrometeorological Institute in Zagreb, which owned the building, with their consent in principle.

Due to the fact that two families of meteorologists lived in the building, their housing had to wait, so in 1979 the "Istra" Astronomical Society of Pula was given the use of only two auxiliary rooms on the eastern side of the building.



Pulski astronomi u svojoj zvjezdarnici

PULA (M. U.) — Astronomsko društvo »Istra«, koje djeluje od kraja 1873. a po mnogo čemu je među najaktivnijima u zemlji, uselilo je konačno u zgradu stare Zvjezdarnice u parku Rudera Boškovića.

Već sljedećeg tjedna, nakon uređenja prostorija, mladi astronomi prenijet će u zvjezdarnicu instrumente i početi ostvarivati opsežni program istraživanja i praćenja svemirskih pojava s mjesta gdje je prije stotinu godina otkriven planetoid »Istra« i još osamdesetak drugih planetoida.

In 1982, the Observatory was handed over to the Astronomical Society "Istra" Pula.

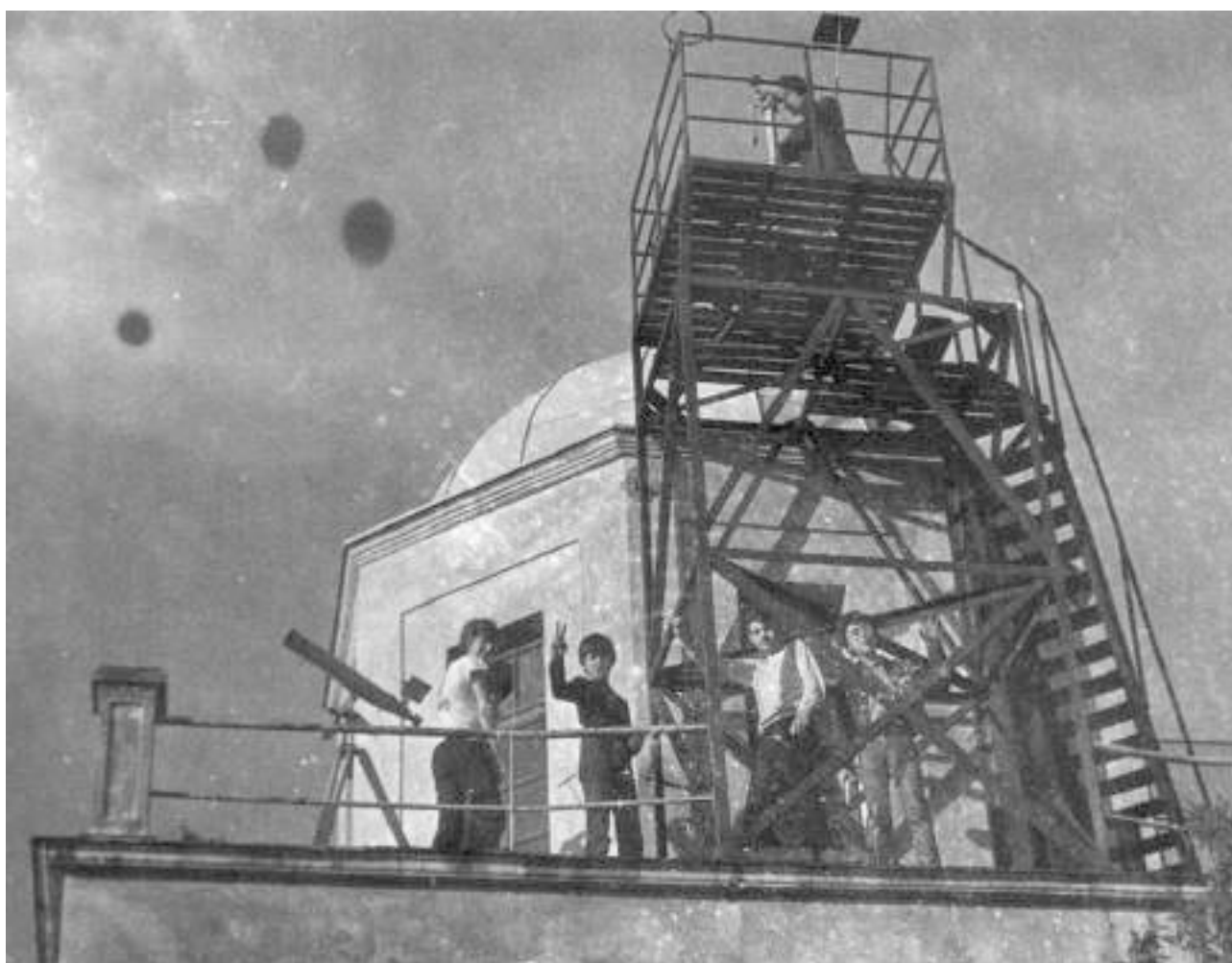
Since the building was in a very dilapidated and neglected state, the members of the society immediately took on the job and initiated the action of its decoration. With the help of work organizations and the great voluntary work of the members of the Society, the Observatory was gradually brought into usable condition, room by room. As the dome was inaccessible at that time, observations were made from the nearby meadows on Monte Zaro, next to the fountain, and partially from the terrace and meteorological scaffolding.

In 1983, at the proposal of the "Istra" Astronomical Society, the Pula Observatory was declared a category C cultural monument by the decision of the Rijeka Regional Institute for the Protection of Monuments (Rijeka, May 30, 1983 under number 435).



The observatory brought additional enthusiasm to the members of the Society, who at that time devoted themselves to astrophotography as a method of observation, primarily observations of the Sun.

In addition to astrophotographic ones, visual observations of the Moon and planets, as well as large meteor streams (mostly Perseids and Orionids, since the weather conditions usually made it impossible to observe the Geminids and Quadrantids) were regularly carried out.





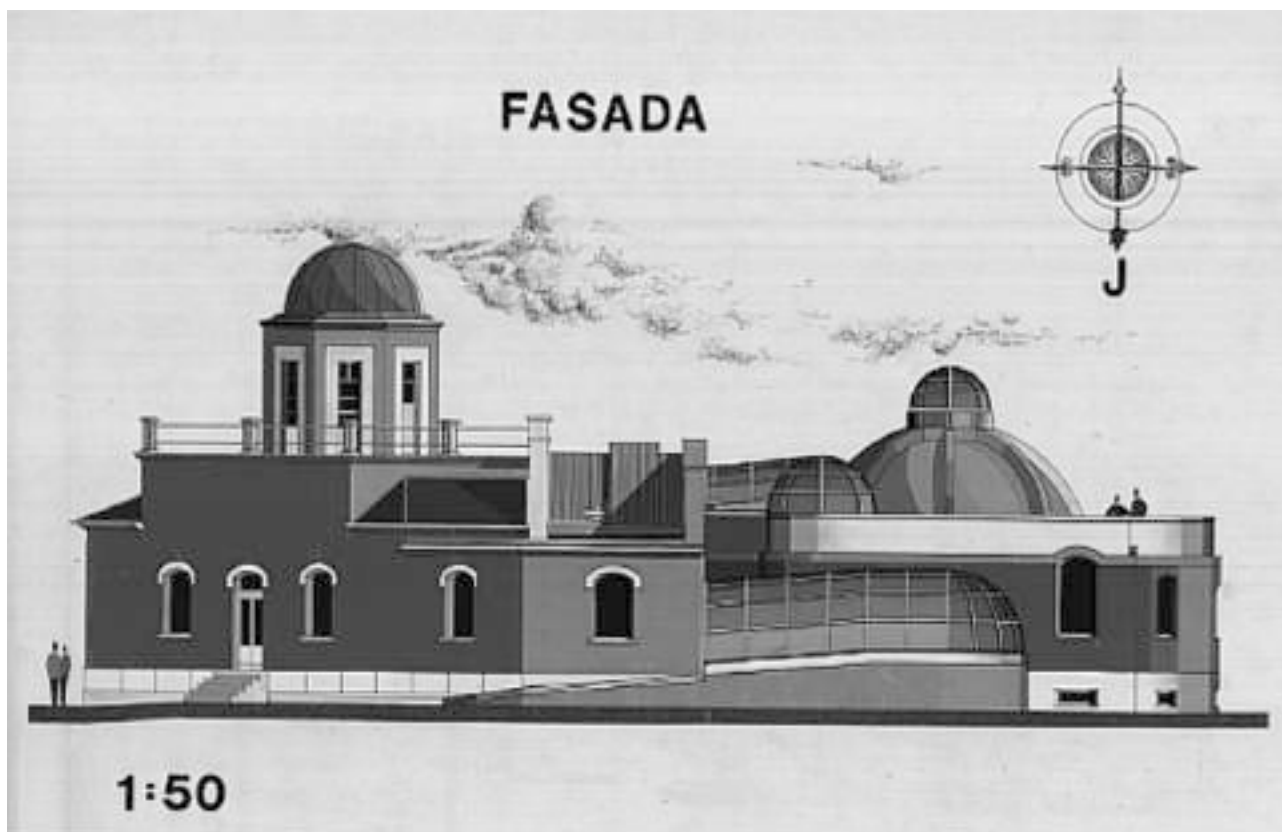
Astronomy school competitions were also held at the Observatory as part of the "Science for the Young" movement. Every year, a large number of primary and secondary school students entered the competition. At the astronomy competition, each student had to present their practical work in addition to the written part of the work. Sometimes these were observations of meteors, planets or variable stars, and sometimes models or advanced works such as research into the efficiency of using the Sun's energy.





Competitions of the "Science for Youth" movement brought the most successful as a final prize participation in an astronomy camp on the island of Prvić, where young lovers of astronomy socialized through learning and observations.





In 1985, a member of the society, Zdenko Kovač, made a project for the renovation and extension of the Observatory with a planetarium as his graduation thesis for his architecture studies. The work grew into a project that was accepted as such, but due to political turmoil and war events, it fell into oblivion.

In 1991, (then still a member, future secretary) Edi Puh published the book "Observatory in Pula: 1871-1991." (materials for the history of Istria)», in which he presented historical facts about the Observatory in a simple and comprehensible manner. The book is a reference point for all those who intend to delve deeper into the details of the history of the Observatory.





The time of the Homeland War largely stopped the work of the "Istra" Pula Astronomical Society. Thanks to the exceptional efforts of individuals (primarily Edi Puh and Lorenzo Putigna, then secretary and chairman), the Observatory was saved from further deterioration. Thanks to the help of the shipyard "Uljanik", in 1999 access to the main dome was finally arranged and its functionality in terms of astronomical observations was restored, and the Observatory could once again receive guests.





The Astronomical Society "Istra" in Pula monitored all events in the sky that are important to citizens, including the partial eclipse of the Sun on August 11, 1999, at the Observatory. Almost all active members of the Association helped and explained to the curious what kind of phenomenon it was. We estimate that several thousand people passed by and watched the event with us that day at Monte Zaro.





Although the members of the Society constantly maintained the Observatory in accordance with the possibilities and funds they received for this purpose from the city of Pula, it was not enough for them to be successful in this. Therefore, at the beginning of 2006, the city of Pula initiated a general rehabilitation of the building, through four phases - in the first phase, the roof over the central part of the Observatory building was repaired.





The second phase of the renovation, carried out in 2007, completed the eastern part of the terrace, which could not be used until that moment. This gave the Observatory a very valuable space where in the future outdoor lectures took place, as well as an observation area for citizens. Thanks to a donation from a telecommunications company, the Astronomical Society "Istra" Pula bought a Celestron 11 telescope and installed it in the main dome of the Observatory.





The third phase of renovation carried out in 2008 covered the roofs around the dome, which definitively solved the leakage problem after many years.

Work has also begun on the southern facade of the Observatory, which used to be a partition wall. The entrance represented communication between the central building and the wing with the domes.





The final stage of renovation in 2009 included the facades, which were demanding in many respects. In order to preserve the original form of the building as much as possible, all available details were reconstructed. It is important to emphasize that the meridian dome is not included in the rehabilitation. The observatory took its final form at the beginning of May 2009, and the opening of the facility was marked by an appropriate event.





After the restoration, the work with the young generations is renewed and continues with the popularization of astronomy in the form of observations and lectures for citizens.

A large number of different groups of students visit the Observatory, where they get information about its history and significance for the development of astronomy in the world.





In some cases, the groups were more numerous than a hundred participants, which required additional organization of visits by members of the Association. Special attention was paid to younger ages, primary school students who, in addition to historical facts about the Observatory in Pula, were able to learn the basics of the motion of the bodies of the Solar System through an interactive presentation.





In 2010, the Astronomical Society "Istra" Pula received the Charter of the Croatian Community of Technical Culture, for its many years of outstanding contribution to the development and improvement of technical culture and astronomy, as well as its contribution to the preservation of cultural monuments - Observatory in Pula

Members of the "Istra" Pula Astronomical Society participate in and support other projects of a similar nature that popularize science. As part of such projects, they go out into nature and conduct observations of the Sun with the necessary safety measures, as well as education in terms of the dangers of looking directly at the Sun.





Young members of the Association at the Observatory were preparing their works for the astronomy competition, as part of the Education Agency. Practical works were presented at a joint lecture, in the presence of parents.

After the renovation in 2009, the number of students who participated in the competitions increased sharply, and their results each year were enough for at least a few of them from school and county competitions to continue to the end, to national competitions where some of them won first places.





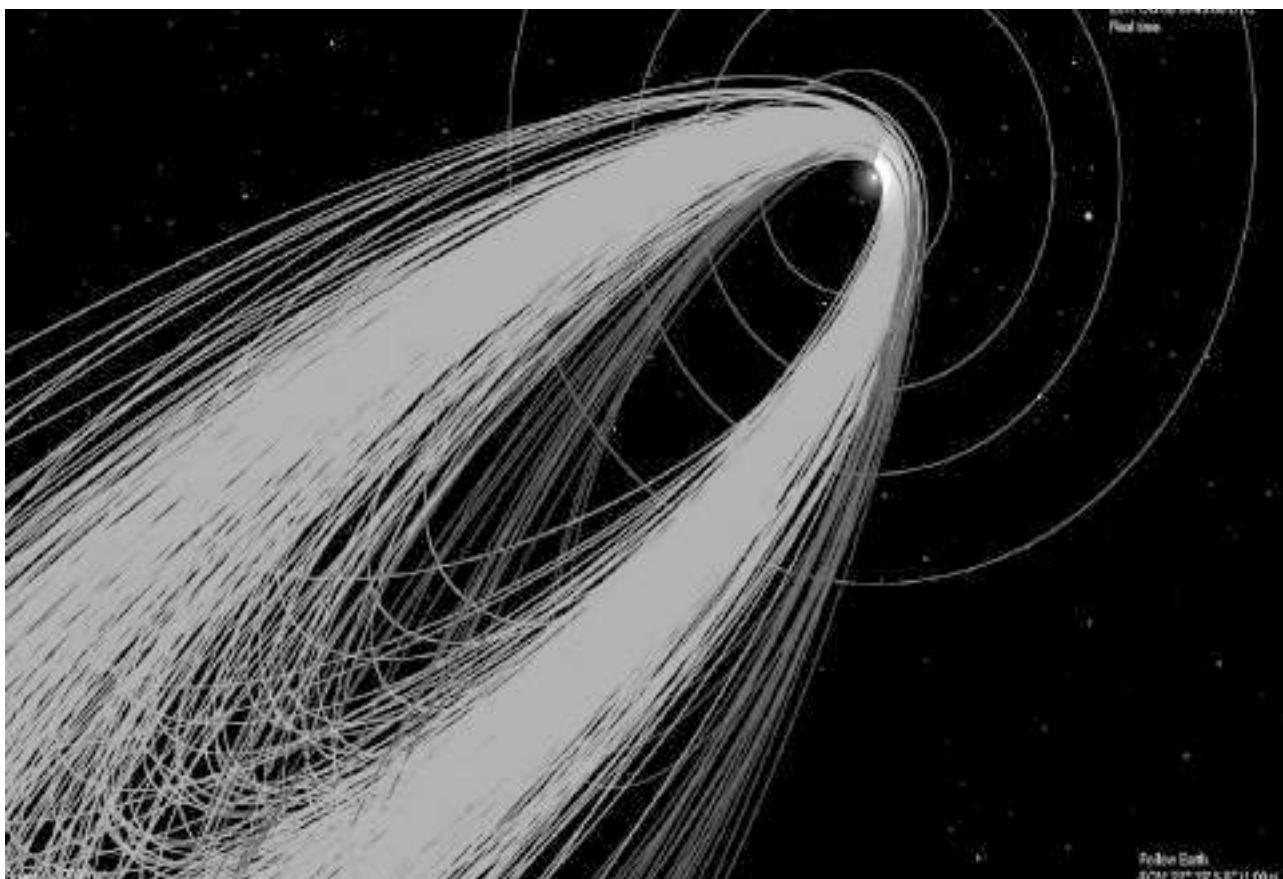
Even while the rehabilitation was in progress, during the maximum of the Perseid meteor shower in 2006, video cameras were tested for the purpose of meteor observations. This experiment proved to be very successful, and the Croatian Meteor Network project resulted from it, in which more than 40 associations and individuals participated throughout Croatia and beyond (several cameras were located in neighboring countries).

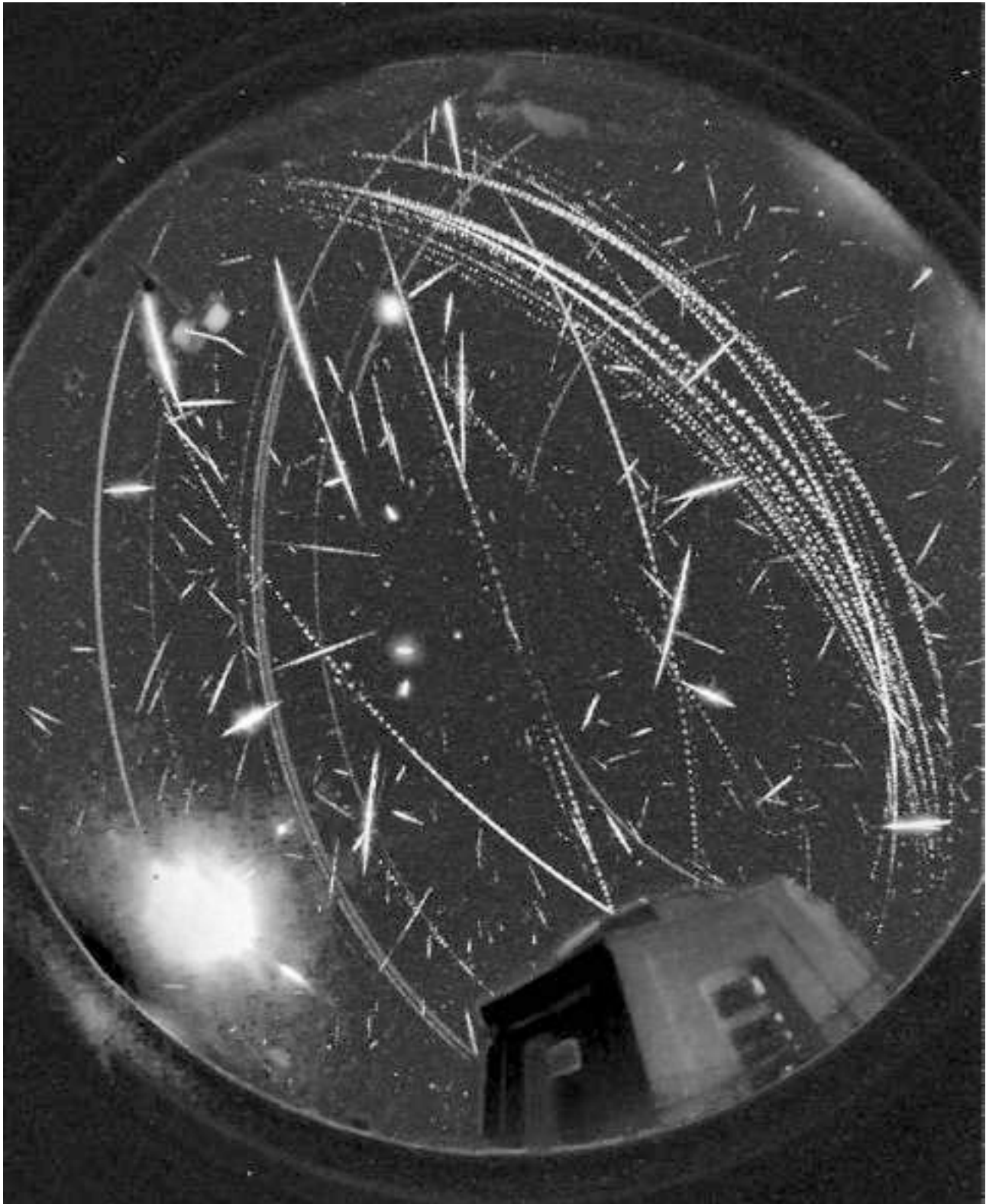


In addition to meteors, the cameras also successfully recorded lightning strikes, as well as rarely observed transient effects in the Earth's atmosphere, so-called "sprites" (Croatian translation: "fireworks"). It turned out that the project of the Croatian Meteor Network was successful beyond all expectations. In cooperation with the Scientific Education Center Višnja, the project resulted in a large number of young astronomers who, with their own solutions, improved the more or less standard methods of observing meteors. As a result, as of 2018 there is a Global Meteor Network, which in 2021 exceeded the number of 500 cameras active worldwide.

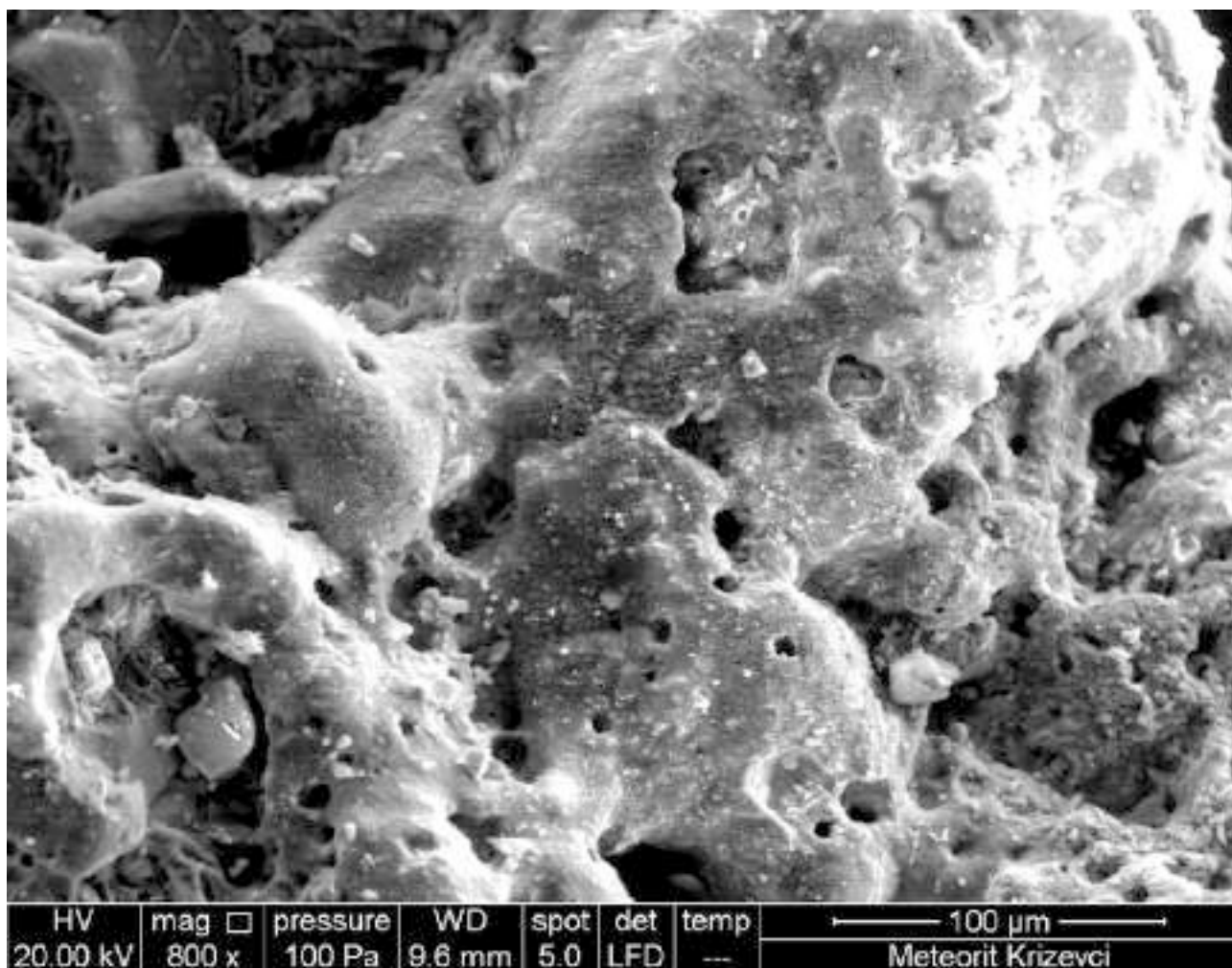


Thanks to the work on meteor astronomy, members of the Association actively participated in several editions of the International Meteor Conference. At these conferences, they presented their results, some of which were related to the discovery of new meteor streams. In total, for now, the Croatian Meteor Network has been recognized for the discovery of 8 new meteor streams, but we expect that number to be significantly exceeded in the near future.





All-sky video of the night of the Geminid stream meteor observation in 2020. In the video in front of the meteor, we see a lot of airplanes, satellites, since the cameras are capable of recording almost everything that the human eye can see. In addition, these cameras are very sensitive in the infrared part of the spectrum, where meteors radiate a significant part of the energy, and the human eye cannot perceive almost anything in that part of the spectrum. In addition to the all-sky camera, the Observatory also has a camera installed to observe telescopic meteors, those that are too faint to be seen. Due to the drastically smaller field of view of that camera, the number of observed meteors is significantly lower.



Members of the "Istra" Pula Astronomical Society take an active part in the "Science Festival" project every year. In 2016, in cooperation with "Metris" from Pula, they presented the analysis of the "Križevci" meteorite, in the discovery of which we had a very important part.

The Association also participates in joint projects of the Croatian Astronomical Union, such as the topic "Women in Astronomy", through appropriate lectures or organized observations.





In order to popularize astronomy and the activities of the Association, members hold lectures at various meetings of amateur astronomers, always striving to spread knowledge about the history of the Observatory.

In addition to lectures as part of astronomers' meetings, members of the Association hold lectures about the Observatory and its history at various gatherings on similar topics.





The Global Meteor Network – Methodology and first results

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The Covid-19 pandemic stopped the activities of the members of the Association towards citizenship. However, the work on meteor observations continues continuously, in addition to observations, scientific papers based on own observations are also published. It should be emphasized that meteor observations, among other things, from the Observatory in Pula are used by the world's leading astronomical organizations, including NASA.

What makes us especially happy is that the growing number of young members, who are now already on the advanced path to becoming engineers, masters and doctors of science, have remained in contact with the Astronomical Society "Istra" Pula, and are very happy to respond to the actions organized by the Association. We hope that this will be the case in the future as well.



Thank you!

We would like to thank everyone who made this publication possible:

Croatian Association of Technical Culture

To the Historical and Maritime Museum of Istria, for providing photographs from the reign of the Austro-Hungarian Monarchy

Edi Puh, Luciano Gržinić and Igor Karasi for the provided materials

To all members of the Astronomical Society "Istra" Pula for their suggestions and dedicated work





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